ECON 100B: Macroeconomics

Class 11: Growth and Ideas 3 (Jones Chapter 6)

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February 24, 2026

Agenda

- Announcements
- Current events
- Review the “old school” Romer model in Chapter 6
  - We’re studying the version in 1e-5e. This will be on the midterm exam
  - (In 6e, the model is a little different, but many punchlines are similar)
- Wrap-up of growth
  - Outline of Solow-Romer model (not on the exam)
  - Growth accounting results
  - Fall 2010 midterm Q3

BERKELEY STUDENT LEARNING CENTER

ECONOMICS PROGRAM

Mastery through Collaborative Learning

ECON 100B

César Chávez Student Center

Monday - Thursday 12 - 5 PM

Service Schedule

Announcements

- Problem Set 2 is due on Gradescope next Friday 2/27
- In-class Midterm the next Thursday 3/5
- Take-home Midterm Essay released 3/8, due 3/20
- Spring Break 3/23-3/27

Current events

- In a 6-3 ruling, the Supreme Court on Friday struck down the U.S. administration's broad use of tariffs under IEEPA

- Three conservatives struck them down because the U.S. Constitution invests the authority to tax in Congress

- Three liberals struck them down because they interpreted IEEPA as failing to grant emergency power to levy tariffs

- Francis (WSJ 2/20/26) "'The American System Was Tested and It Works' Says Toy Maker in Tariff Case"
  - "[The case] was about how we make decisions as a country about the imposition of taxes…. What makes our country **investible** is respect for the **rule of law**."

Alysa Liu, raised in Richmond, trained in Oakland, attends UCLA, now Olympic gold!

First half of ECON 100B

- Chapter 2 shows how we <u>measure</u> income, Y = C + I + G + NX

- **Chapters 3–6 discuss <u>long-run growth</u>**

    - Chapter 3 introduced the growth rate rules

$$g[X \times Y] = g[X] + g[Y] \qquad g[X/Y] = g[X] - g[Y]$$

$$g[X^a \times Y^b] = a \cdot g[X] + b \cdot g[Y]$$

    - Chapter 4 introduces Cobb-Douglas production

$$Y = A\ K^{1/3}L^{2/3}$$

with constant returns to scale
and declining marginal productivity of K or L

    - Chapter 5 develops the Solow Model

    - **Chapter 6 develops the Romer Model**, in which scientists produce A

Learning objectives in Chapter 6

- Goods and services are produced by **ideas**, capital, and labor
- Ideas are special
  - They are **nonrival** and can be used in many places at the same time
  - New, useful ideas might be scarce, but existing ideas are not scarce
  - Ideas can still be **excludable**, legally kept from use by others
- If ideas can also be produced, they help us achieve increasing returns to scale and growth in income per person
- To produce ideas, we probably need
  - An **ideas sector** with labor
  - Some kind of tool to make ideas excludable, like patent and copyright law

The “old school” Romer model

- In ECON 100B, we will learn and use the Romer model appearing in 1e through 5e of Chad Jones’s *Macroeconomics*

- In 6e, a different version appears in §6.3
  - It’s a little harder
  - To me, its benefits don’t seem to outweigh its costs

- Interested students are welcome to look at the new version

The Romer model of growth and ideas

- Like before, suppose there is one production good **Y**
- Let **A** be the stock of ideas, and new ideas are produced just like Y
- For simplicity, we'll forget about capital and imagine that **labor** is the only factor of production
- Labor is rival — people can either work in goods (Y) production OR in idea production (A) but not both
- But ideas are **nonrival** — ideas can work in goods (Y) production and in idea production (A) at the same time

An aside: Stocks of factor inputs produce flows of production through production functions

- The output of a production function is a **flow** of something new, like new cars or new potato chips

- Production functions take as inputs some **stocks** of productive factors, like technology, capital, and labor

- Both of these production functions produce a flow as a function of stocks

$$Y = F(A, K, L) = AK^{1/3}L^{2/3}$$

a function of capital & labor

$$\Delta L = \Phi(M, D)$$

a “fertility function” of moms & dads

- In both cases, the factor inputs are stock variables and the outputs are flow variables

Setup of the "old school" Romer model from Jones 1e-5e

- In the goods sector, "food" is a function of ideas and farmer labor L_{yt}. There is no capital, and food rises linearly with farmers

$$Y_t = A_t \; L_{yt}$$

- In the ideas sector, **new ideas** ΔA_t are a function of current ideas and scientist labor L_{at}. There is no capital, and new ideas rise linearly with scientists, whose productivity rises with $\bar{z}$

$$\Delta A_t = \bar{z} \; A_t \; L_{at}$$

- Total labor $\bar{N}$ is split between farm and science. Science share is ℓ

$$L_{at} + L_{yt} = \bar{N} \qquad\qquad L_{at} = \ell \; \bar{N}$$

Remarks

$$Y_t = A_t \; L_{yt} \qquad\qquad \Delta A_t = \bar{z} \; A_t \; L_{at}$$

- Ideas are very important, but they also a **means to an end**. The end is utility from consumption, and we can't consume ideas
- Income or "food" per person equals $y_t \equiv \frac{Y_t}{N}$
- More ideas help the economy produce more food, but they come at a resource cost of L_{at} workers who could otherwise be farming
- The magic is that ideas A_t can be used simultaneously in food production and in new idea production. Ideas are nonrival inputs

Solving the model, step 1

- What does our model say about income per person, which is now $Y/\bar{N}$ in this framework?

$$Y_t = A_t \; L_{yt} \qquad \text{and} \qquad L_{yt} = (1 - \ell)\bar{N}$$

$$Y_t = A_t \; (1 - \ell)\bar{N}$$

$$\frac{Y}{\bar{N}} \equiv y_t = A_t \; (1 - \ell)$$

- Income per person is a function of the total stock of knowledge
- A new idea increases everyone's income because it's nonrival
- By contrast, in the Solow model, income per person depends on capital per person, not the level of capital overall
- This is because capital is a rival good in production. Two workers cannot use the same capital good at the same time

Solving the model, step 2

- Ideas are important. What can we say about growth in ideas?

$$\Delta A_t = \bar{z}\ A_t\ L_{at} \qquad \text{and} \qquad L_{at} = \ell\ \bar{N}$$

$$\Delta A_t = \bar{z}\ A_t\ \ell\ \bar{N}$$ now divide by A_t

$$\frac{\Delta A_t}{A_t} = \bar{z}\ \ell\ \bar{N}$$

- The growth rate of the stock of ideas, $g[A]$, is constant over time and proportional to the number of idea producers in the economy

- Since the growth rate of A is constant at $g[A] = \bar{z}\ \ell\ \bar{N}$, we can write:

$$A_t = A_0 \left(1 + g[A]\right)^t$$

where A_0 is the initial level of ideas

Solving the model, step 3

$$A_t = A_0 \left(1 + g[A]\right)^t$$

- If this is a formula for the stock of ideas A_t then recall

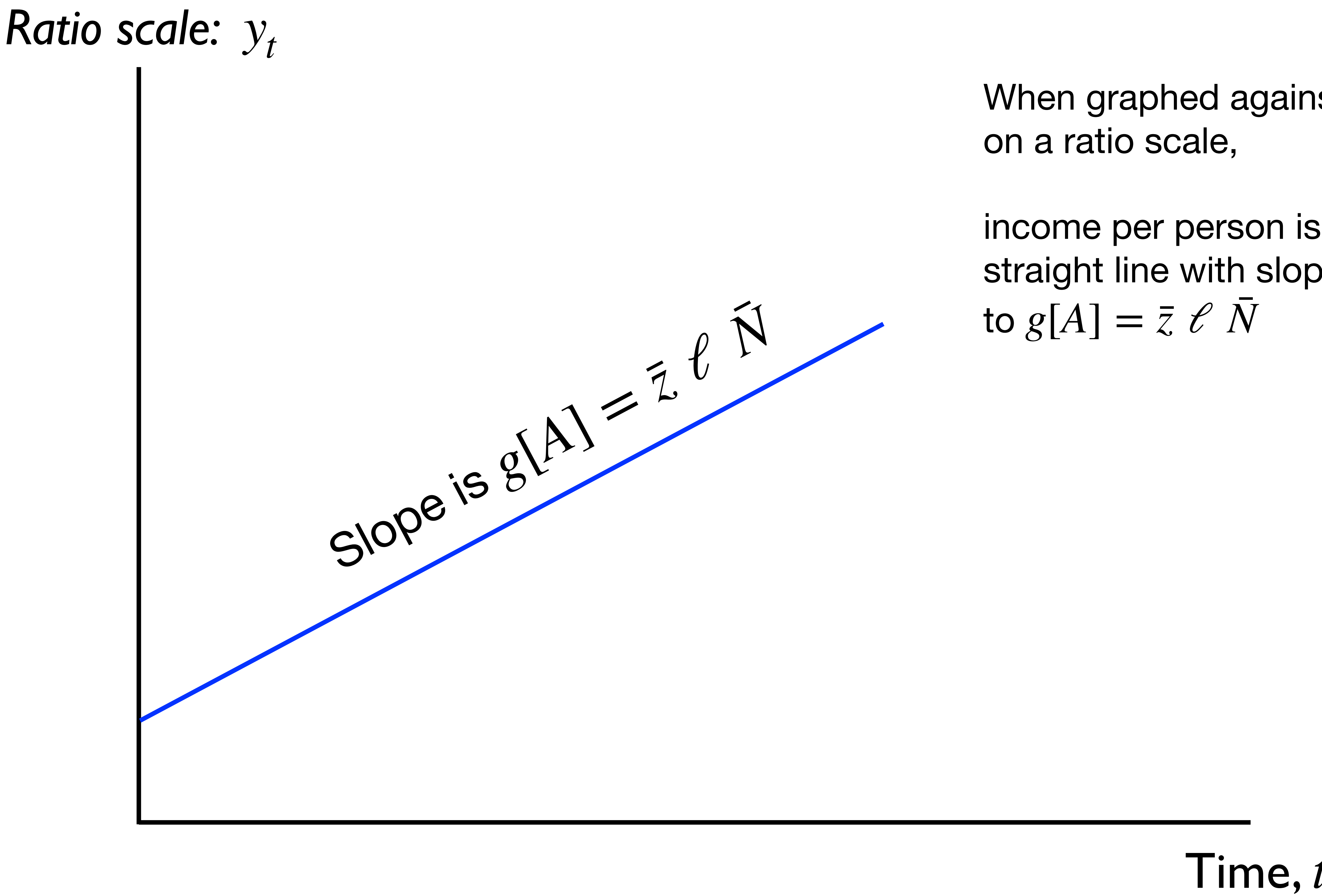
$$\frac{Y}{\bar{N}} = y_t = A_t \ (1 - \ell) \qquad \text{substitute for } A_t$$

$$y_t = A_0 \left(1 + g[A]\right)^t \ (1 - \ell) \qquad \text{or rearranging,}$$

$$y_t = A_0 \ (1 - \ell)\left(1 + g[A]\right)^t$$

- This is exponential growth in income per capita
- No other model has produced sustained growth over time
- Here the "steady state" is really a **"balanced growth path"**

$$y_t = A_0\,(1-\ell)\big(1+g[A]\big)^t$$



When graphed against time on a ratio scale,

income per person is a straight line with slope equal to $g[A] = \bar{z}\ \ell\ \bar{N}$

Why is there growth in the Romer Model?

- Here, ideas are the engine — in the Solow model, it was capital
- Consider the idea accumulation equation:

$$\Delta A_t = \bar{z}\ A_t\ L_{at}$$

- There are constant returns to ideas in idea production:
  - The exponent on A_t equals one
  - As we gain knowledge, the return to knowledge does not fall
- Capital accumulation in the Solow Model is different!

$$\Delta K = s\ AK^{1/3}L^{2/3} - dK$$

- There are diminishing returns to capital in capital production, because $F(A, K, L)$ exhibits diminishing returns in capital
- Why the difference? Ideas are nonrival, capital is rival

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What else affects growth in the Romer Model?

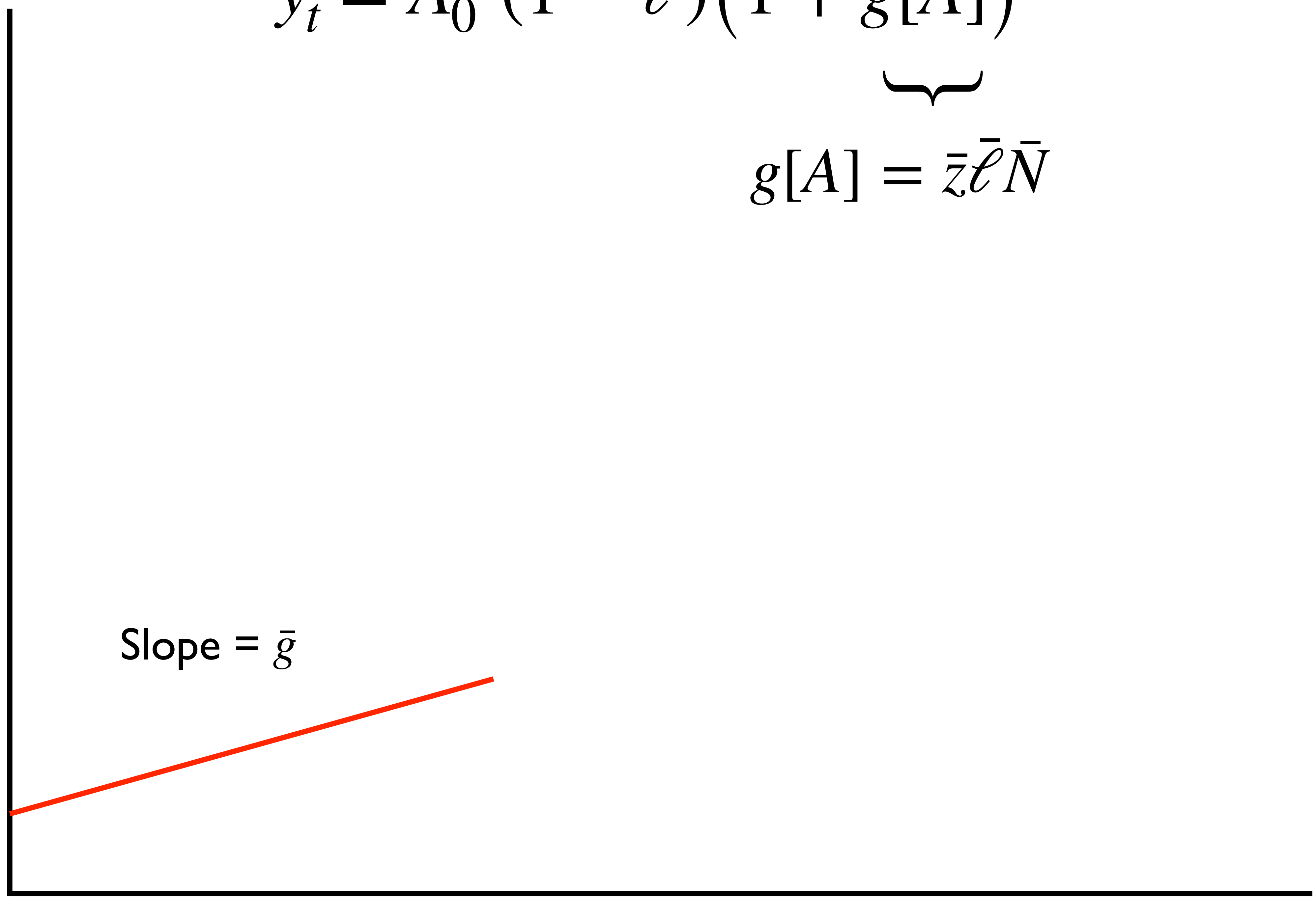
- In this model, there are 4 parameters:
    - $\bar{N}$ the size of the population
    - ℓ the share of the population in "science" or ideas
    - $\bar{z}$ the productivity of scientists
    - A_0 the starting level of ideas

- What happens when we change these parameters?
- In the "old school" Romer Model, there is a balanced growth path and no transition dynamics

Suppose $\bar{N}$ increases, what happens to growth in y_t?

Ratio scale: y_t

$$y_t = A_0\,(1-\ell)\Big(1+\underbrace{g[A]}_{g[A] = \bar{z}\bar{\ell}\bar{N}}\Big)^t$$



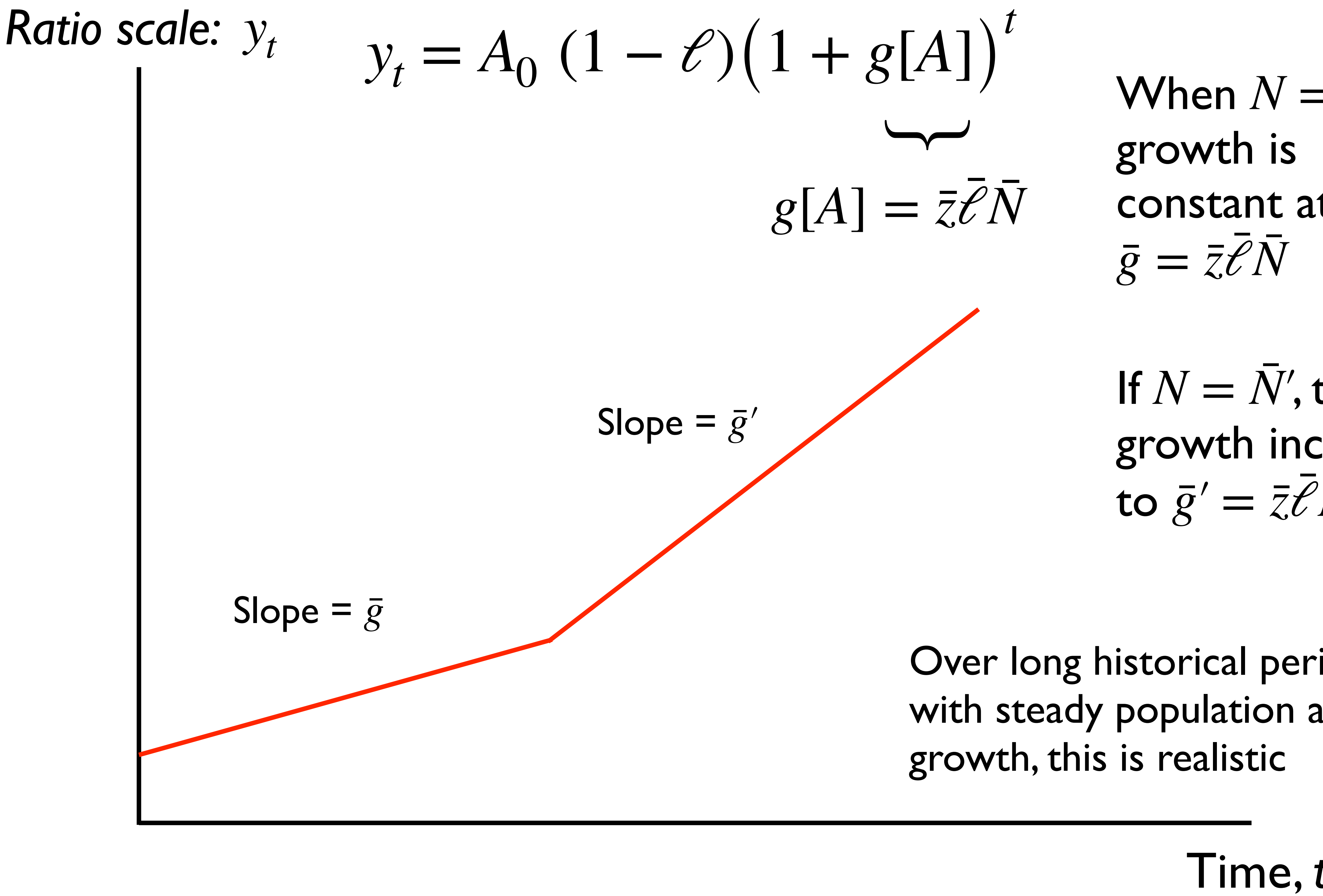
Time, t

Suppose $\bar{N}$ increases, what happens to growth in y_t?

$$y_t = A_0\,(1-\ell)\underbrace{\big(1+g[A]\big)^t}_{g[A]\,=\,\bar{z}\bar{\ell}\bar{N}}$$

When $N = \bar{N}$, growth is constant at $\bar{g} = \bar{z}\bar{\ell}\bar{N}$

If $N = \bar{N}'$, then growth increases to $\bar{g}' = \bar{z}\bar{\ell}\bar{N}'$

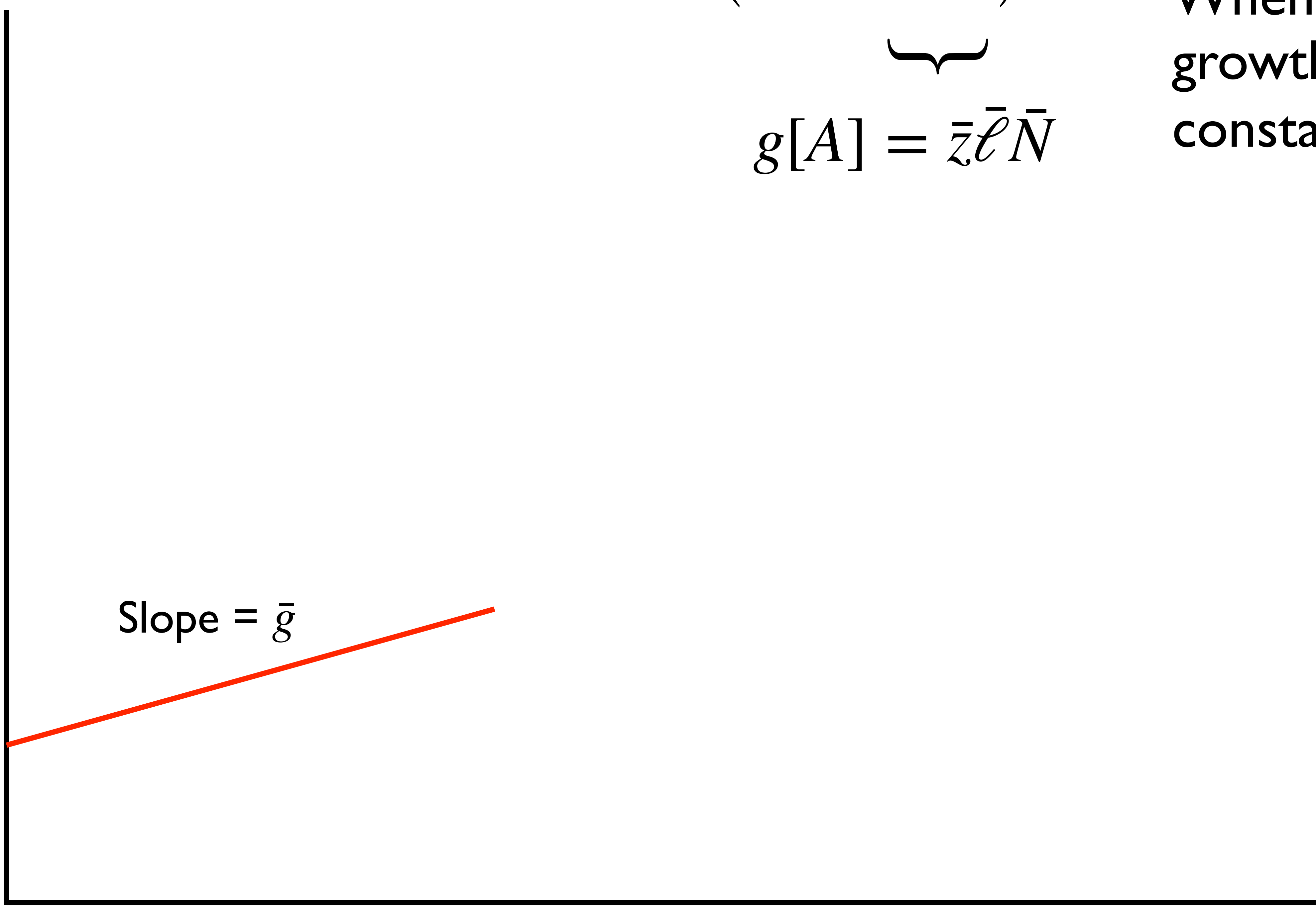


Over long historical periods, with steady population and no growth, this is realistic

Suppose $\bar{\ell}$, the labor share of scientists increases

$$y_t = A_0\,(1-\ell)\big(1+\underbrace{g[A]}_{g[A]=\bar{z}\bar{\ell}\bar{N}}\big)^t$$

When $\ell = \bar{\ell}$, growth is constant at $\bar{g}$



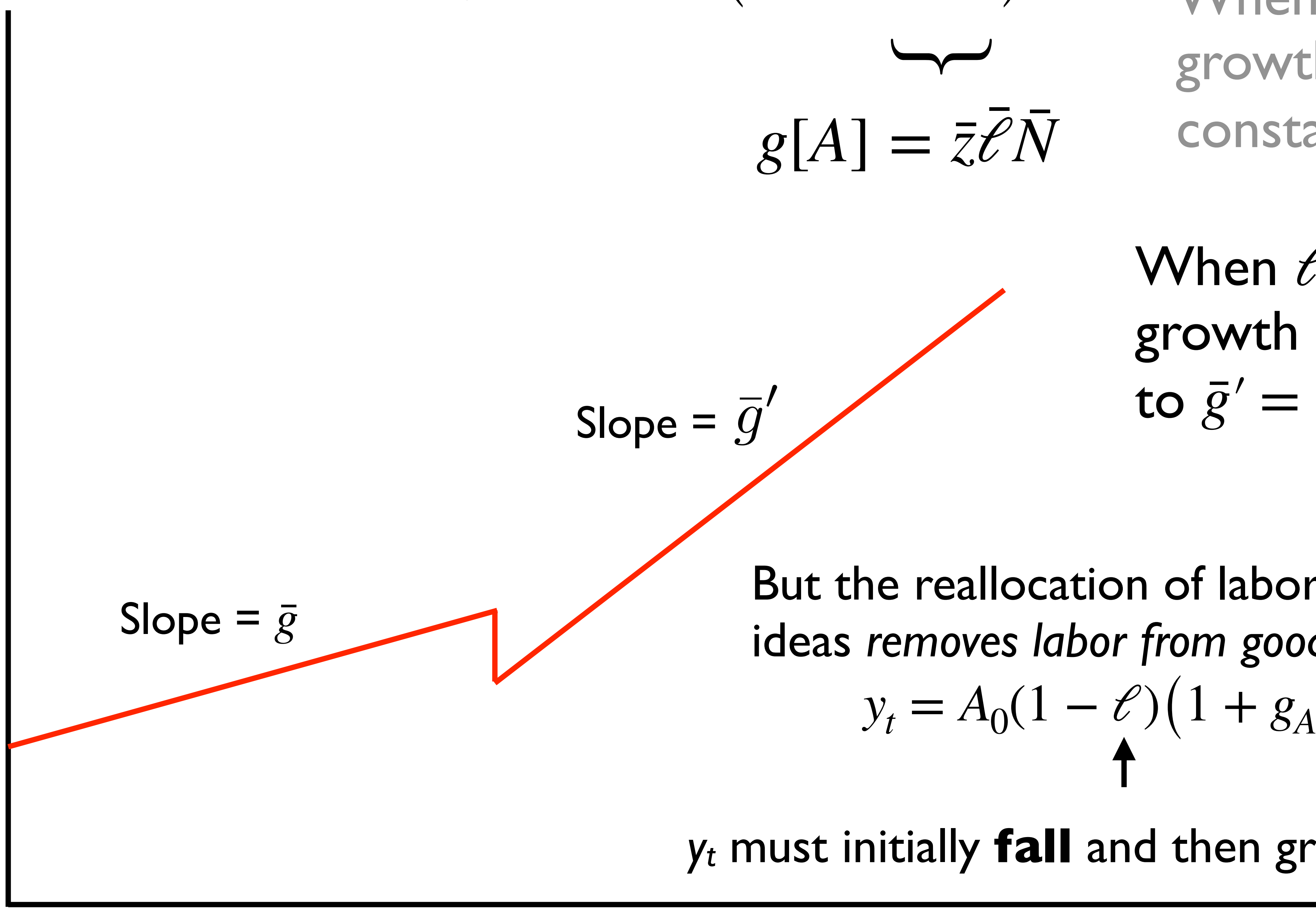
Suppose $\bar{\ell}$, the labor share of scientists increases

Ratio scale: y_t

$$y_t = A_0\,(1 - \ell)\big(1 + \underbrace{g[A]}_{g[A] = \bar{z}\bar{\ell}\bar{N}}\big)^t$$

When $\ell = \bar{\ell}$, growth is constant at $\bar{g}$

When $\ell = \bar{\ell}' > \bar{\ell}$ growth increases to $\bar{g}' = \bar{z}\bar{\ell}'\bar{N}$



But the reallocation of labor toward ideas *removes labor from goods production:*

$$y_t = A_0(1 - \underset{\uparrow}{\ell})\big(1 + g_A\big)^t$$

y_t must initially **fall** and then grow faster

Time, *t*

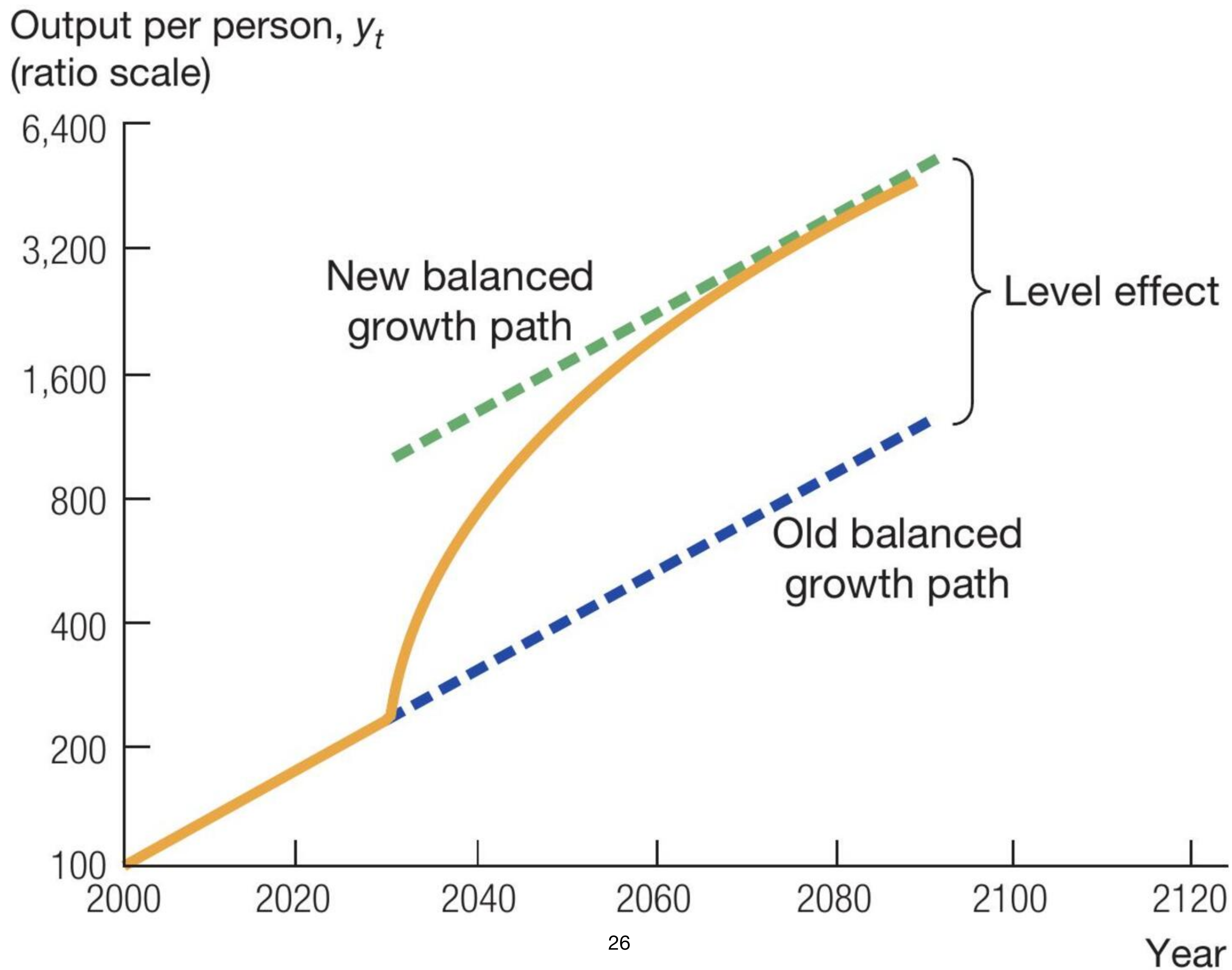
Reality is more like Solow-Romer

“Somer” or “Rolow” maybe?

- The Solow model explains some things pretty well
  - Why some countries grow at different rates for decades at a time based on how far they are from their steady states
  - For example, postwar Japan and Germany; recently Korea, which has much higher saving/investment than other SE Asian countries
- The Romer model explains the source of long-run growth in Y/L
- We can combine these two models for a complete view of growth

[THIS IS NOT ON THE EXAMS]

In the Solow-Romer model in §6.9, an increase in the saving rate pushes the economy up to a higher balanced growth path



Growth Accounting in §6.5

How much growth has been attributable
to growth in physical factors
versus growth in TFP?

$$Y_t = A_t K_t^{1/3} L_{yt}^{2/3}$$

Take growth rates of both sides

$$g_{Y_t} = g_{A_t} + \frac{1}{3} g_{K_t} + \frac{2}{3} g_{L_{yt}}$$

Now subtract g_{Lt}

$$\underbrace{g_{Y_t} - g_{L_t}} = \underbrace{g_{A_t}} + \frac{1}{3}\underbrace{\left(g_{K_t} - g_{L_t}\right)} + \frac{2}{3}\underbrace{\left(g_{L_{yt}} - g_{L_t}\right)}$$

Growth in income per person = Growth in Total Factor Productivity + Growth in capital per person + Growth in hours worked per person

With this equation, we can decompose growth over time in income per person into growth in its components

Growth in these components of GDP per person has changed over time in the U.S. — how they have changed tells us about the sources of growth

U.S. growth due to:	1948–2002	1948–1973	1973–1995	1995–2002
Income per person	2.5%	3.3	1.5	3
Capital per person	0.9	0.9	0.7	1.3
Hours worked per person	0.2	0.2	0.2	0.4
Total Factor Productivity	1.4	2.1	0.6	1.2

Growth accounting has produced some lively debates, for example about the sustainability of growth in Singapore and other Asian "Tigers"

Singaporean growth due to:	1966–1990	1966–1970	1970–1980	1980–1990
Income per person	4.2	7.6	3.8	3.3
Capital per person	3.4	4	4.3	2.4
Hours worked per person	0.6	-1.1	0.4	1.5
Total Factor Productivity	0.2	4.6	-0.9	-0.5

Source: Young, QJE 1995

Fall 2010 Midterm

Andrew Jacobs (NYT 10/6/2010) "Rampant Fraud Threat to China's Brisk Ascent"

3. (25 points) *The economic costs of plagiarism.*

   An October 7 article in the *New York Times* stated the following:

   > [A] lack of integrity among researchers is hindering China's potential and harming collaboration between Chinese scholars and their international counterparts.... [T]he results of a recent government study ... [reveal that] a third of the 6,000 scientists at six of the nation's top institutions admitted they had engaged in plagiarism or the outright fabrication of research data.... "If we don't change our ways, we will be excluded from the global academic community," said Zhang Ming, a professor of international relations at Renmin University in Beijing. "We need to focus on seeking truth."

   What are the costs of this type of activity?

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Consider China as a Romer economy, with knowledge (A_t) and goods (Y_t) production, productivity of scientists equal to $\bar{z}$, and total population fixed at $\bar{N}$ and employed in either knowledge (share $\bar{\ell}$) or goods production:

$$\begin{aligned} \Delta A_t &= \bar{z} A_t L_{at} && (4) \\ Y_t &= A_t L_{yt} && (5) \\ \bar{N} &= L_{at} + L_{yt} && (6) \\ L_{at} &= \bar{\ell} \bar{N} && (7) \end{aligned}$$

(a) (3 points) Rearrange equation (4) and use (7) to find the growth rate of A_t, g_A, in terms of $\bar{z}$, $\bar{\ell}$, and $\bar{N}$. Show your work!

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(a) (3 points) Rearrange equation (4) and use (7) to find the growth rate of A_t, g_A, in terms of $\bar{z}$, $\bar{\ell}$, and $\bar{N}$. Show your work!

Answer:

$$\begin{aligned}
\Delta A_t &= \bar{z} A_t L_{at}\\
\Delta A_t &= \bar{z} A_t \bar{\ell}\bar{N}\\
\frac{\Delta A_t}{A_t} &= \frac{\bar{z} A_t \bar{\ell}\bar{N}}{A_t}\\
g_A &= \bar{z}\bar{\ell}\bar{N}
\end{aligned}$$

(b) (3 points) Algebra tells us that income per capita in this model is

$$y_t = A_t(1 - \bar{\ell}) \tag{8}$$

Substitute for A_t using your knowledge about something that has a growth rate equal to g_A. Rewrite it to find y_t as a function of $\bar{A}_0$, $\bar{z}$, $\bar{\ell}$, and $\bar{N}$.

8

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Answer:

$$A_t = A_0(1 + g_A)^t$$

$$y_t = A_0(1 + \bar{z}\bar{\ell}\bar{N})^t(1 - \bar{\ell})$$

(c) (3 points) Look at the first paragraph of the citation and Prof. Zhang's quote. What parameter in the Romer model will change if plagiarism or fabrication of data does not stop, and why? (Hint: Of what use is collaboration among researchers?)

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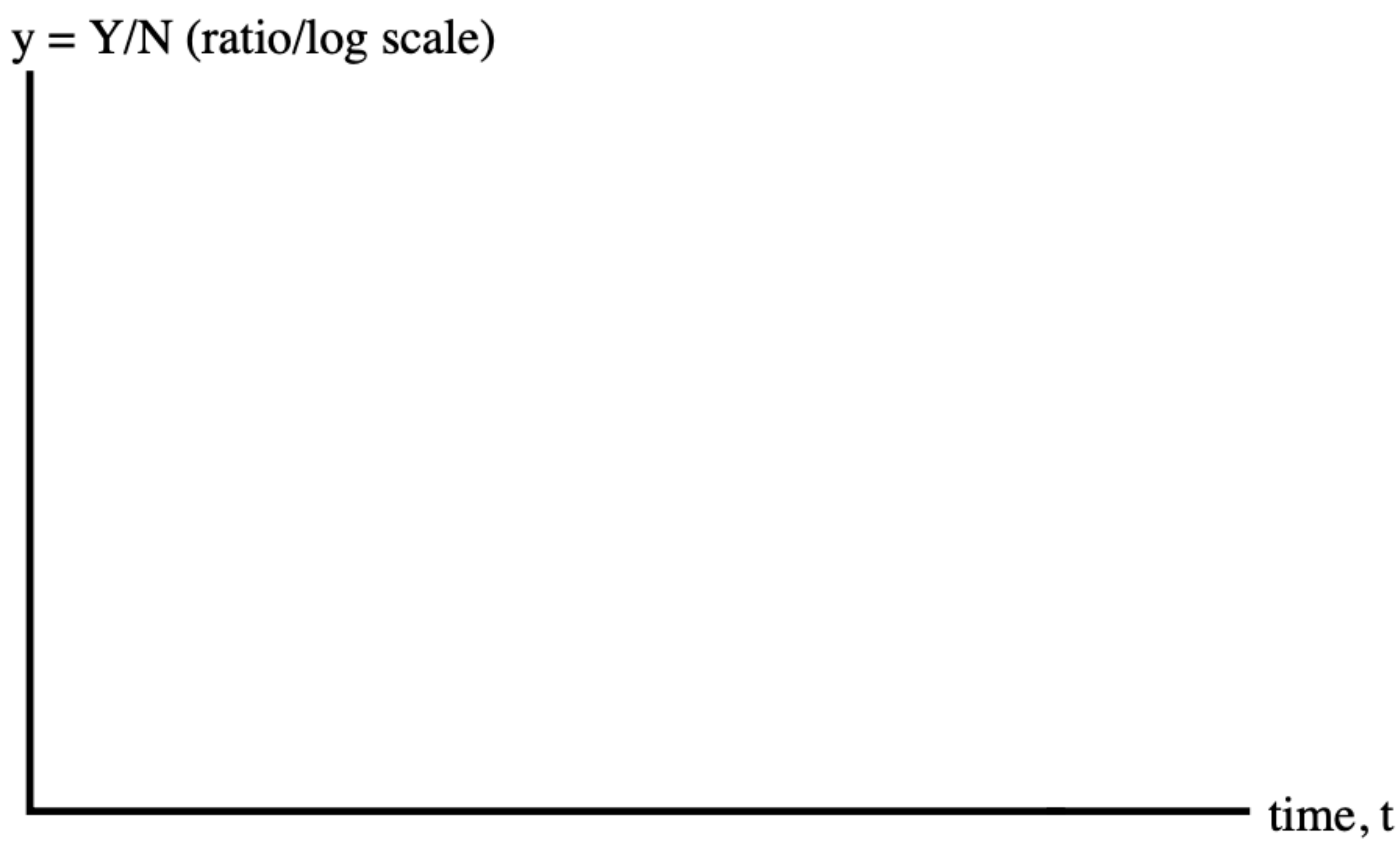
Answer: The first paragraph of the quote argues that faking research "is hindering . . . potential and harming collaboration. . . ." Zhang warns of being excluded from the international community. Both of these are statements about the **research productivity of scientists** in China, or the $\bar{z}$ parameter, and specifically about its falling lower, maybe to zero. Why have scientists at all if all they do is steal other people's work and have zero net productivity of their own!

(d) (4 points) Illustrate the effect that this change would have on the Chinese economy in the Romer Model using the axes below. Label clearly the time at which this happens, and describe what you have drawn. Be mindful of changes in slope versus changes in level.



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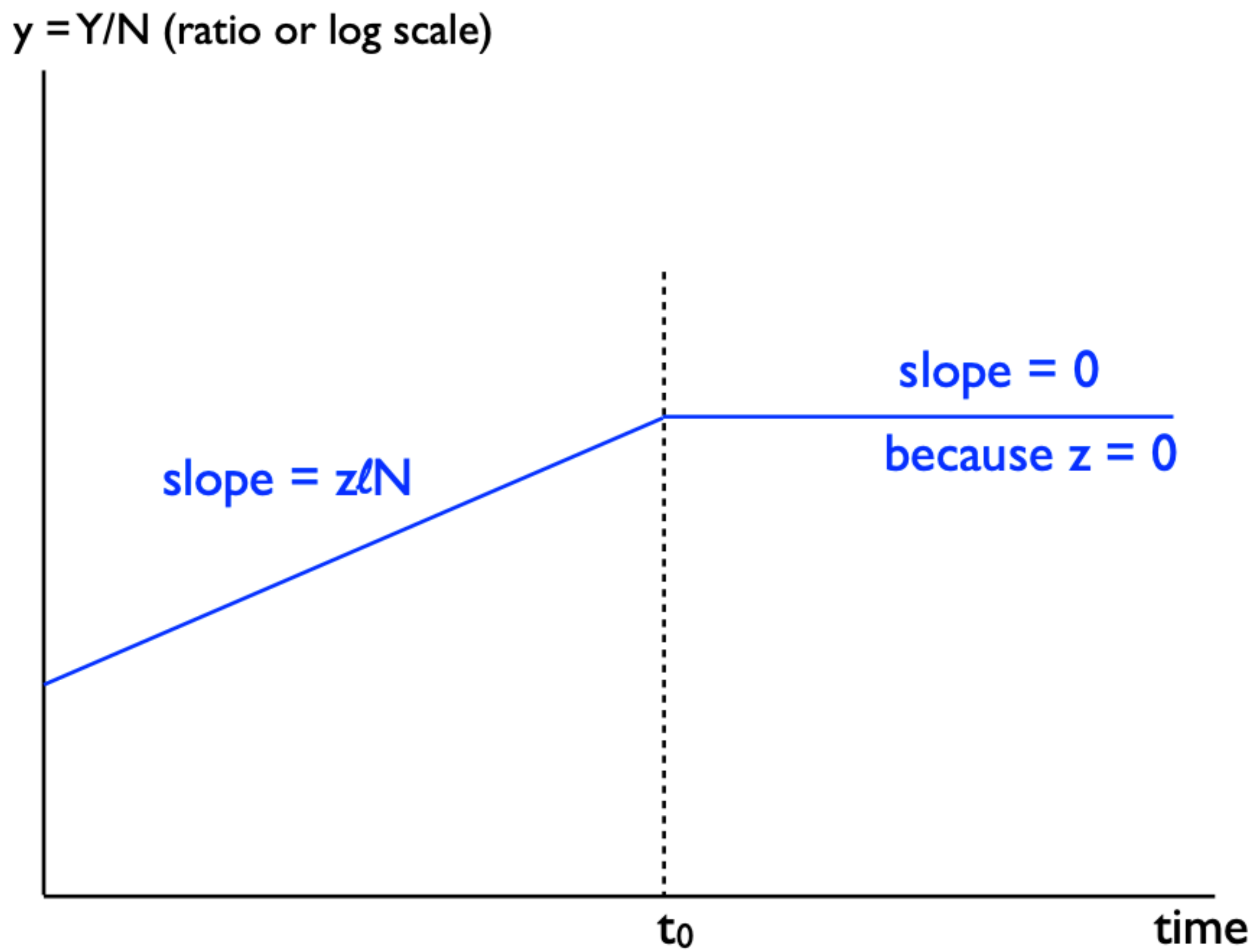

Answer: If Chinese scientists were to find that falsification of results and plagiarism resulted in hindered potential, harmed collaboration, and exclusion from international scientific communities, the research productivity of scientists, $\bar{z}$, would probably fall to zero. This reduces growth to zero whenever it happens, say at time t_0. At that time, the level of income per person is preserved with no further growth.

In the Romer Model, the marginal product of scientists can be written:

$$MPL_a = \bar{z}A_t$$

(e) (4 points) Make an argument for why Chinese scientists themselves should stop plagiarizing.

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(e) (4 points) Make an argument for why Chinese scientists themselves should stop plagiarizing.

Answer: Chinese scientists are paid their marginal product. If their productivity is zero, their marginal product and hence their wage should be zero. Also, there is no growth in their wage, but this is rather secondary. Still, either answer would be acceptable.

The marginal product of labor that is devoted to goods production (think "farmers") is

$$MPL_y = A_t$$

(f) (4 points) Would Chinese goods producers prefer to hide the embarrassing prevalence of plagiarism and falsification among Chinese scientists, or would they prefer to expose it in hopes of changing it?

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(f) (4 points) Would Chinese goods producers prefer to hide the embarrassing prevalence of plagiarism and falsification among Chinese scientists, or would they prefer to expose it in hopes of changing it?

Answer: From a purely economic standpoint, Chinese goods producers (i.e., everyone else) would prefer to expose plagiarism and falsification among Chinese scientists if doing so stopped the problem and resulted in continued research productivity. The reason is that goods producers' wages are dependent on the level of ideas, and if falsification results in a halt to growth in ideas, goods producers would not be happy.

(g) (4 points) Would changing China's one-child policy help the situation in any way? Does it at all depend on what Chinese scientist parents teach their Chinese scientist kids about plagiarism?

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Answer: In the Romer Model, a larger population will also include more scientists, and that produces faster growth in income per person because more scientists produce more ideas, which fuel growth. But the present problem isn't really related to the number of scientists at all. No matter how many there are, if they all plagiarize and falsify results, their collective research productivity will be zero.

If on the other hand new scientists grow up learning new rules governing scientific pursuits, such as how not to plagiarize and falsify results, the story might change. One way of guaranteeing future scientific productivity might be focusing on training new generations of scientists, much of which is done by moms and dads, who themselves might be scientists. (Or they might not be. It's probably better if they aren't!)